D. Monster Game

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: binary search, sortings, two pointers | https://codeforces.com/problemset/problem/2193/D

You have been gifted a new game called "Elatyh". In the game, you are given n swords, each with its own strength. In particular, the sword numbered i has a strength of a_i. The game consists of n levels, each of which features a monster.

You start at level 1 and progress further. To pass level i and move on to level $i + 1$, you need to defeat the monster at level i. To defeat the monster at level i, you need to deal it b_i sword strikes. The swords in the game are very fragile, so they can only deal **one** strike before breaking. If you complete level n or run out of swords, you can finish the game and proceed to score calculation.

Before the game, you are allowed to choose the difficulty level. If you choose difficulty x, swords with a strength less than x will not affect the monsters. The game score in this case is equal to x multiplied by the number of levels completed. Your task is to choose the game difficulty in such a way as to maximize the game score.

Input

Each test consists of several test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The following describes the test cases.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$).

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le n$).

It is guaranteed that the sum of the values of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum game score.

Example

Input

```
5
3
1 3 4
2 1 1
2
2 3
1 1
4
1 2 3 4
2 2 1 1
6
4 4 1 4 5 4
2 2 4 1 2 2
10
1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000
1000000000 1000000000
1 1 1 1 1 1 1 1 1 1
```

Output

```
3
4
3
8
10000000000
```

Note

Consider the first test case. Optimal difficulty to choose is 3. If difficulty is 3, you can deal strikes with swords number 2 and 3. With 2 swords you can complete 1 level, so the game score is 3· 1 = 3.